

BLAST: Basic Local Alignment Search Tool

BIFX-550 · Class 06 | S. Ravichandran, Ph.D.
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Before we begin this week's lecture:

- Discussion on last week Labs
- GitHub setup
- Discussion of the choice of BLAST DB

Session Agenda

- **Warm-up: Sequence Alignment Concepts (brief review & discussion)**
- **Part 1 — What is BLAST and why do we need it?**
 - Homologs, databases, and the search problem at scale
- **Part 2 — BLAST flavors**
 - blastp, blastn, blastx, tblastn, tblastx — when to use each
- **Part 3 — How BLAST works (3 phases)**
 - Phase 1: Word compilation; Phase 2: Ungapped + gapped extension; Phase 3: Traceback
- **Part 4 — E-values, scores, and search strategies**
 - Mathematical relationship between E-value and alignment score
- **Biostar Handbook Part XVII · Chapters 82–85 · Computer Lab: NCBI BLAST**

Warm-up: Alignment Concepts

What governs a 'best' alignment?

What Governs a Sequence Alignment?

- **There is no single 'universally best' alignment — the result depends on:**
 - Scoring matrix (BLOSUM62, PAM250, etc.)
 - Gap open and extension penalties
 - Algorithm choice: global (Needleman–Wunsch) vs. local (Smith–Waterman)
- **Concrete example — three alignments of GATTACA vs. GATCA:**
 - Align1: score 15.0 (affine gap, end-gap correction applied)
 - Align2: GATTACA / GATCA-- score 4.5 (no end-gap correction)
 - Align3: GATTACA / GA-T-CA score 5.0
- **Key takeaway: always report the matrix, gap costs, and algorithm used**
- *When sequences are similar, most algorithms agree. When diverged, parameter choices matter enormously.*

Align1

GATTACA
GAT-CA-

Align2

GATTACA
GATCA—

Align3

GATTACA
GA-T-CA

Similarity vs. Identity — and When to Use Protein vs. DNA

- **% Identity:** fraction of positions that are exactly the same residue
- **% Similarity:** fraction that are identical OR biochemically similar (e.g., Leu↔Ile, positive score in BLOSUM)
 - Example — CAL vs. C-I: C is identical; A and I are dissimilar (negative BLOSUM score); L and I are similar
- **Why protein for distant homologs?**
 - 20-letter alphabet carries more information per position than 4-letter DNA
 - Synonymous (silent) DNA substitutions obscure functional conservation
 - Protein comparisons reach ~1 BYA; DNA comparisons max out ~600 MYA
- **Use BLASTX or TBLASTN when you only have a DNA sequence but want protein-level sensitivity**

Part 1: What Is BLAST and Why Do We Need It?

The database search problem at scale

The Database Search Problem

- **Goal: find all sequences in a database that are similar (likely homologous) to a query**
- **Scale of the problem (NCBI nr protein database):**
 - ~171 million sequences (Sep 2018) → ~620 million sequences (Sep 2023)
 - A naïve Smith–Waterman search over the entire database is $O(m \times N)$ — prohibitively slow
- **BLAST (Altschul et al., 1990) solves this with a heuristic three-phase strategy:**
 - Phase 1 — Compile a word index of high-scoring short sequence segments from the query
 - Phase 2 — Scan the database for those words, then extend matches into alignments
 - Phase 3 — Traceback to produce final gapped alignments with statistical scores
- *Trade-off: BLAST is not guaranteed to find the single best alignment (heuristic), but it is extremely fast and sensitive in practice*

What Is BLAST Used For?

- **Identifying homologs**
 - Find orthologs across species and paralogs within a genome
 - Important caveat: homology $\neq$ guaranteed same function; similarity $\neq$ homology
- **Identifying an unknown sequence**
 - Paste a mystery sequence $\rightarrow$ SMART BLAST returns functional annotation
- **Organism classification / taxonomic ID**
 - Match 16S rRNA or COI barcoding sequences against taxonomic databases
- **Discovering new genes (Find-a-Gene project)**
 - TBLASTN: use a known protein to search unannotated genomic or EST sequences
 - Genomic context: Genomic BLAST shows the chromosomal neighborhood of a hit

Part 2: Flavors of BLAST

Choosing the right tool for your query and database

BLAST Flavors — Query × Database Combinations

Program	Query	Database	Notes / Best Use
blastp	Protein	Protein	Standard protein homology search; uses BLOSUM62 by default
blastn	Nucleotide	Nucleotide	DNA vs. DNA; exact word match (default word size 11); fast
blastx	Nucleotide	Protein	Translates query in all 6 frames; detects protein-coding regions in DNA
tblastn	Protein	Nucleotide	Translates database on-the-fly; find unannotated homologs — key for Find-a-Gene
tblastx	Nucleotide	Nucleotide	Translates both query and DB; most sensitive but slowest; not recommended for large DBs

BLAST2 = direct comparison of two sequences (no database scan)

BLAST Flavors — Query × Database Combinations

Query type and database type determine the program and how many translated comparisons are made

Program	Query	Searches	Database	Translated
blastp standard homology search	protein	1	protein	none
blastn DNA against DNA	DNA	1	DNA	none
blastx identify what a contig encodes	DNA	6	protein	query
tblastn Find-a-Gene, Step 2	protein	6	DNA	database
tblastx most computationally intensive	DNA	36	DNA	both

Searches = protein-level comparisons per database sequence. Concept after Pevsner, Bioinformatics and Functional Genomics, 3rd ed.; redrawn for BIFX-550.

BLAST2 = direct comparison of two sequences (no database scan)

BLASTX: Translating DNA Before Searching

- **Why 6 frames?**
- Double-stranded DNA can be read in 3 forward frames and 3 reverse-complement frames
- An unannotated DNA contig may contain a protein-coding region in any of these frames
- **What BLASTX does:**
 - Translates the nucleotide query in all 6 reading frames
 - Searches each translated protein against a protein database (e.g., nr)
 - Reports the frame and strand of the best-matching protein hit
- **Use BLASTX when you have a transcript/EST and want to identify the encoded protein**

Selecting a Database

- **For protein queries (blastp / tblastn):**
 - nr — non-redundant: merges GenBank CDS translations, PDB, SwissProt, PIR, PRF; ~620M seqs (2023)
 - RefSeq — curated, non-redundant reference sequences; ~281M proteins (2023); prefer for well-studied organisms
 - PDB — structural sequences only; use when you want a structural hit
- **For nucleotide queries (blastn / blastx):**
 - nr/nt — includes GenBank, EMBL, DDBJ, PDB, RefSeq; does NOT include EST, STS, WGS, GSS, TSA, Patents, HTGS
 - RefSeq_RNA / RefSeq_Select — smaller, higher quality; good for paralog searches
 - Organism filter: restrict by taxid (e.g., taxid:9721 for whales/dolphins) to reduce noise
- **nr is large and noisy; start with RefSeq when possible, expand to nr if you find too few hits**

Part 3: How BLAST Works — 3 Phases

Compile → Extend → Traceback

Pha

Entries for the BLOSUM62 matrix at a scale of $\ln(2)/2.0$.

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V	B	J	Z	X	*
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0	-2	-1	-1	-1	-4
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3	-1	-2	0	-1	-4
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3	4	-3	0	-1	-4
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3	4	-3	1	-1	-4
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1	-3	-1	-3	-1	-4
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2	0	-2	4	-1	-4
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2	1	-3	4	-1	-4
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3	-1	-4	-2	-1	-4
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3	0	-3	0	-1	-4
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3	-3	3	-3	-1	-4
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1	-4	3	-3	-1	-4
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2	0	-3	1	-1	-4
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1	-3	2	-1	-1	-4
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1	-3	0	-3	-1	-4
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2	-2	-3	-1	-1	-4
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2	0	-2	0	-1	-4
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0	-1	-1	-1	-1	-4
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3	-4	-2	-2	-1	-4
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1	-3	-1	-2	-1	-4
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4	-3	2	-2	-1	-4
B	-2	-1	4	4	-3	0	1	-1	0	-3	-4	0	-3	-3	-2	0	-1	-4	-3	-3	4	-3	0	-1	-4
J	-1	-2	-3	-3	-1	-2	-3	-4	-3	3	3	-3	2	0	-3	-2	-1	-2	-1	2	-3	3	-3	-1	-4
Z	-1	0	0	1	-3	4	4	-2	0	-3	-3	1	-1	-3	-1	0	-1	-2	-2	-2	0	-3	4	-1	-4
X	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-4
*	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	1

the

e →

Phase 1 Example: Word Scoring with BLOSUM62

Query fragment: S A Y I D P A A F N D
(4) (4) (7) (4) (6) (7) (4) (4) (6) (6) (6) ← BLOSUM diagonal scores

Sliding window, word size = 3, threshold $T = 15$:

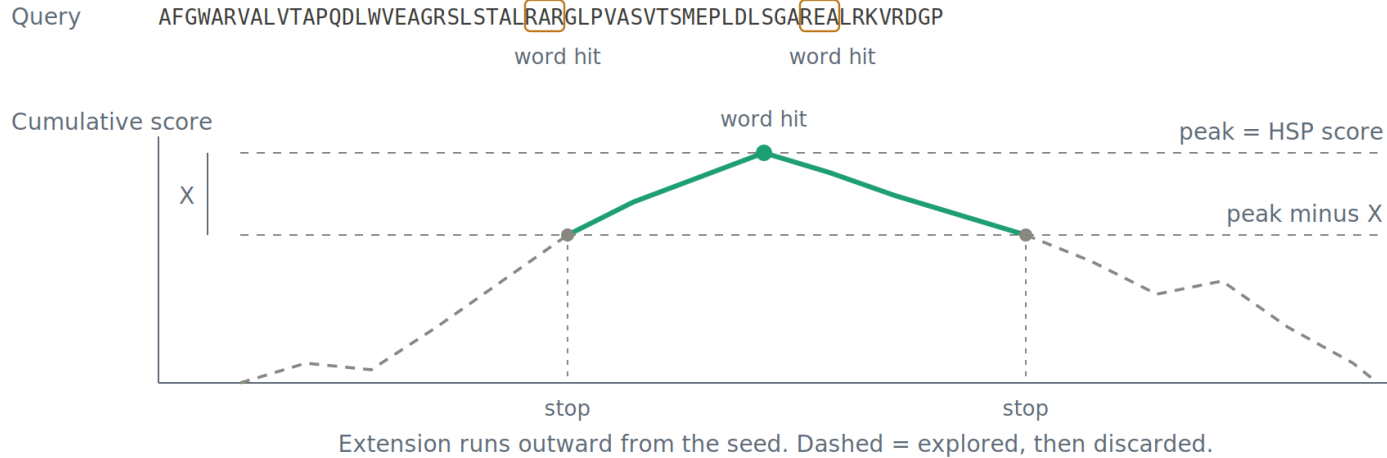
Word	Composition scores	Total	Keep?
SAY	S (4) + A (4) + Y (7)	= 15	✓ (exactly at threshold)
AYI	A (4) + Y (7) + I (4)	= 15	✓
YID	Y (7) + I (4) + D (6)	= 17	✓
IDP	I (4) + D (6) + P (7)	= 17	✓
DPA	D (6) + P (7) + A (4)	= 17	✓
PAA	P (7) + A (4) + A (4)	= 15	✓
AAF	A (4) + A (4) + F (6)	= 14	✗ (below threshold)
AFN	A (4) + F (6) + N (6)	= 16	✓
FND	F (6) + N (6) + D (6)	= 18	✓

Words scoring $\geq T$ are loaded into the hash-table index. Lower T = more words = higher sensitivity but slower.

Phase 2: Ungapped Extension, Then Gapped Extension

- **Database scan:**
 - Look up each database position in the hash table — $O(1)$ per position
 - A hit requires two word matches in the same database sequence within a window (two-hit model) → filters out accidental single-word matches
- **Ungapped extension (first):**
 - Extend the match left and right without allowing gaps; accumulate score
 - Stop when the score drops more than X below the running maximum (X -drop criterion)
 - Only alignments above score threshold S advance to the next step
- **Gapped extension (second):**
 - Apply a full Smith–Waterman-style DP with affine gap penalties to the HSP region
 - Compute the High-Scoring Segment Pair (HSP) score
- *The X -drop and two-hit filter together give BLAST its speed — most database positions are eliminated at Phase 1*

Phase 2: Extension Illustrated



Resulting alignment after gapped extension

Query AFGWARVALVTAPQDLWEAGRSLSTALRARGLPVASVTSMEPLDLSGAREALRKVRDGP
 |||||+||||| |||||+||||| ||||| ||||| |||||+|||||
 Sbjct AFGWARIALVTA - - - LWVEAGKSLSTALRARGLPVLVTSMEPSDLSGAREAMRKVRDGP

Vertical bar = identical, plus = similar (positive score), dash = gap

HSPs above threshold S advance to Phase 3; others are discarded.

Phase 3: Traceback and Final Alignment

- **Input: all HSPs that survived the Phase 2 score threshold**
- **Steps:**
 - Identify gap positions (INDELs) from the gapped extension DP matrix
 - If applicable (blastp, tblastn), apply composition-based statistics adjustment to correct for amino acid composition bias in the query or hit
 - Produce the final gapped alignment with match/mismatch/gap characters
 - Calculate the raw score S , bit score S' , and E-value for each alignment
- **Output:**
 - Graphical overview of hits, then individual alignments sorted by E-value
 - Each alignment reports: score, E-value, % identity, % positives (similarity), % gaps, alignment length
- **Phase 3 is where BLAST becomes statistically rigorous — raw speed comes from Phases 1–2**

Part 4: E-Values, Scores & Search Strategies

The mathematics behind BLAST output

The E-Value: Definition and Intuition

- **Definition (Altschul):**

- *"The number of different alignments with score $\geq S$ expected to occur by chance in a database search"*
- In other words: $E = 1$ means you'd expect one random match this good by chance.

- **Key properties:**

- E-value scales with database size N and query length m : $E \propto m \times N$
- Doubling the database roughly doubles the E-value for the same hit
- E-value is not a p-value, but when $E \leq 0.1$, $E \approx p$ (Poisson approximation)
- Default NCBI threshold: $E = 0.05$ — up to 5 random hits per 100 searches expected
- For short queries (e.g., 5-aa peptide), NCBI automatically raises the threshold (e.g., $E = 200,000$) because short queries cannot achieve high scores
- **Always report E-value alongside the alignment — never interpret a hit without it**

E: “Number of different alignments with scores equal or greater than some score S that are expected to occur in a DB search by chance”

$E \longleftrightarrow P$;
using Poisson

- Prob of finding 0 alignments (or none) with score $\geq S$ is

$$e^{-E}$$

$$p(k \text{ chance alignments with score } \geq S) = \frac{E^k e^{-E}}{k!}$$

The average number of events in an interval is designated as L.

- Prob. of finding at least one alignment with score $\geq S$ is

$$p = 1 - e^{-E}$$

Note, $E(S) = K.N. e^{-\lambda S}$
N = Search space

- This is called “p-value” associated with S.
- When $E \leq 0.1$, $p \approx E$ (almost = to)

The Score–E-Value Relationship

- **Normalized (bit) score:**

- $S' = (\lambda \times S - \ln K) / \ln 2$ where S = raw score, λ and K are matrix-specific constants

- **E-value from bit score:**

- $E = N / 2^{S'}$ where N = search space = $m \times n$ (query length $\times$ total DB length)

- **Interpretation:**

- Every additional bit of score halves the E-value — the relationship is exponential
- High score $\rightarrow$ low E-value $\rightarrow$ statistically significant
- Score inversely proportional to E-value; score proportional to alignment quality

- **Why bit scores rather than p-values?**

- Easier to reason about the number of expected high-scoring alignments (HSAs) than about probability values when scanning hundreds of millions of sequences

Start from the two definitions:

$$E = K \cdot m \cdot n \cdot e^{(-\lambda S)}$$

$$S' = (\lambda S - \ln K) / \ln 2$$

Solve the second for λS :

$$\lambda S = S' \cdot \ln 2 + \ln K$$

Substitute into the exponential:

$$e^{(-\lambda S)} = e^{(-S' \cdot \ln 2)} \cdot e^{(-\ln K)} = 2^{(-S')} \cdot (1/K)$$

Put that back into E:

$$E = K \cdot m \cdot n \cdot 2^{(-S')} / K = m \cdot n \cdot 2^{(-S')}$$

With $N = m \cdot n$ (the search space, query length times total database length):

$$E = N / 2^{S'}$$

Impact of Word Threshold T on Speed vs. Sensitivity

- **Word threshold T controls how many neighborhood words are compiled in Phase 1:**
 - Higher T → fewer words compiled → faster search → may miss distant homologs
 - Lower T → more words compiled → slower search → higher sensitivity
- **Similarly, E-value threshold controls what is reported:**
 - Stringent (e.g., $E = 1 \times 10^{-6}$): fewer, more confident hits; good for well-annotated organisms
 - Permissive (e.g., $E = 10$): more hits; useful when exploring distant homologs
- **Word size:**
 - blastn default = 11; reducing to 7 catches more diverged DNA matches (slower)
 - blastp default = 3; rarely adjusted but can be lowered for very short queries
- **Key principle: parameters optimized for similar sequences should not be blindly applied to diverged ones**

Matrix Choice in BLAST

BLOSUM	% Similarity Range	Typical BLAST Use
BLOSUM90	70–90%	Short, highly similar sequences
BLOSUM80	50–60%	Family member identification
BLOSUM62	30–40%	Default — broadest utility (NCBI BLAST default)
BLOSUM45	< 30%	Weak, long-range / highly diverged alignments
BLOSUM30	< 20%	Extreme divergence; use with caution

Altschul 'Triple Strategy': run default BLOSUM62 + one higher + one lower matrix; compare hits to identify the right range.

Search Strategies

What to do when you have too many — or too few — hits

Too Many Hits? Narrowing the Search

- **Restrict by database:**
 - Switch from nr to RefSeq — smaller, curated, less redundant
 - Filter by organism: enter a common name or NCBI taxid in the 'Organism' box
 - Restrict by region: use the domain of interest (UniProt / CDD domain coordinates)
- **Tighten scoring:**
 - Lower E-value cutoff (e.g., 1×10^{-6} instead of 0.05)
 - Switch to a higher BLOSUM (e.g., BLOSUM80) if your query is closely related to the targets
 - Exclude low-complexity / repetitive regions: enable compositional adjustment or DUST/SEG filter
- **Example — multi-domain query (HIV-1 Pol, 1435 aa):**
 - A long multi-domain protein will hit many partial matches; consider splitting the query by domain

Too Few Hits? Expanding the Search

- **Check for accidental restrictions: reset the BLAST form to clear organism/database limits from a previous run**
- **Expand the database:**
 - Include HTGS, GSS, or WGS databases for unassembled/draft genomic sequences
 - Add model organism sequences if looking for conserved function
- **Relax scoring parameters:**
 - Increase E-value threshold
 - Switch to a lower BLOSUM (e.g., BLOSUM45) or higher PAM for distant homologs
 - Reduce word size (blastn: 7; blastp: 2)
- **Use profile-based methods: (Future Class)**
 - PSI-BLAST: iteratively builds a position-specific scoring matrix from initial hits — dramatically increases sensitivity for distant homologs
 - HMM-based searches (HMMER, InterPro) — covered in later classes

Computer Lab: NCBI BLAST

Biostar Handbook Part XVII, Chapters 82–85

Lab Exercise 1: Nucleotide BLAST (blastn)

- **Exercise from Biostar Handbook — Unannotated transcript from Ryukyu Spiny Rat**
 - Step 1: retrieve the sequence
 - NCBI Nucleotide → search accession GHEE01192317.1 (use the direct URL — main search may not find it)
 - Step 2: run Megablast against RefSeq_Select
 - Expected top hit: Creatine Kinase Muscle (CKM) in mouse, rat, and human
 - Interpretation: this unannotated spiny rat transcript is likely the ortholog of mammalian CKM
 - Step 3: extend the search
 - Query NM_001824.5 (human CKM) with what DB will result in a hit to the Chromosome context?
 - Step 4: restrict to whales/dolphins (taxid: 9721) — do you find CKM in cetaceans?

Lab Exercise 2-3: Paralog Identification

- **Lab Exercise 2: Continuing from the CKM search:**
 - Repeat the blastn search using RefSeq_RNA (a smaller, curated database)
 - Question: do you now see the paralog creatine kinase B (CKB) in the hit list?
 - If yes — what does this tell you about searching nr vs. RefSeq_RNA for paralog detection?
- **Lab Exercise 3: BLAST parameters exploration**
 - Take any protein query of your choice
 - Run blastp three times: BLOSUM62 (default), BLOSUM80, BLOSUM45
 - Record the number of hits and E-value range for each run
- **Discussion questions:**
 - Which matrix gave the most hits? Why?
 - Did any hits appear in BLOSUM45 that were absent from BLOSUM62? Are they biologically plausible?
 - How would you decide which result to trust?

Reading BLAST Output

- **Graphic overview:** horizontal bars show where in the query each subject aligns; color = score range
- **Hit table:** each row is one subject sequence — scan the E-value and % identity columns first
- **Individual alignment — what each field means:**
 - Score (bits): normalized alignment score — higher is better; comparable across different databases
 - Expect (E): expected random hits with this score or better — lower is more significant
 - Identities: exact matches / alignment length (e.g., 163/661 = 25%)
 - Positives: identical + similar (positive BLOSUM score) / alignment length
 - Gaps: number of gap characters / alignment length
 - Query sequence is always on top; subject on the bottom; '|' = identical, '+' = similar, '-' = mismatch
- **Reformatting options:** switch to tabular format for downstream parsing; mismatches shown in pink in the graphical view

Key BLAST Parameters — Quick Reference

- **E-value cutoff:** lower → more stringent (e.g., 1×10^{-6} for confident hits; 10 for exploratory searches)
- **Word size:** lower → more sensitive but slower (blastn default 11 → try 7; blastp default 3 → try 2)
- **Matrix:** BLOSUM62 default; BLOSUM80 for closely related; BLOSUM45 for diverged
- **Gap costs:** match the matrix defaults unless you have a reason to change them
- **Max target sequences:** raise from 100 to 500+ when exploring large families
- **Filters & masking:**
 - Low-complexity filter (DUST for DNA, SEG for proteins): ON by default — prevents hits driven by repetitive sequence
 - Composition-based statistics: adjusts for amino acid bias — recommended for blastp/tblastn
 - Organism / Entrez query: powerful — e.g., 'txid9606[Organism]' restricts to human RefSeq
- *Caution: parameters optimized for one problem (e.g., closely related proteins) should not be blindly reused for a different query*

Session Summary

- **BLAST is a heuristic approximation of Smith–Waterman — fast, but not exhaustive**
 - 3 phases: (1) word index compilation, (2) ungapped + gapped extension, (3) traceback & statistics
- **Flavors: blastp, blastn, blastx, tblastn, tblastx — choose based on query and database types**
- **E-value = expected random hits with score $\geq S$; scales with database size and query length**
 - $E = N / 2^{S'}$ — every extra bit of score halves the E-value
- **Strategies:**
 - Too many hits → restrict DB, raise E cutoff, higher BLOSUM, organism filter
 - Too few hits → expand DB, lower E cutoff, lower BLOSUM, reduce word size, PSI-BLAST
- **Always report: program, database, matrix, gap costs, E-value cutoff, date of search**
- **Next: Multiple Sequence Alignment (MSA) — Biostar Handbook Chapters 7–8**

Find-a-gene Overview

“New Gene” == discovery of some DNA sequence in a DB that has not been annotated yet

Find-a-Gene Tips

